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Current Opinion in Systems Biology

Machine learning for predicting drug–drug interactions: Graph neural networks and beyond

Peter Petschner[1,2,3], Anh Duc Nguyen[2,4], Canh Hao Nguyen[2] and Hiroshi Mamitsuka[2,5]

Identification of interacting drugs before application would be imperative to mitigate the serious risk represented by drug–drug interactions for patient health. Machine learning–based methods are increasingly recognized by regulatory agencies as tools with a central role in drug development, including the identification of novel interactions. In recent years, graph and hypergraph neural networks delivered promising performance improvements compared to non–graph-based methods on the field. In this primer, we discuss recent developments of graph and hypergraph neural networks and highlight the potential of incorporating protein and metabolite data into the identification task to provide a new, more comprehensive, systems biology–based perspective on drug–drug interactions.

Addresses
[1] Department of Pharmacodynamics, Semmelweis University, Nagyvarad Ter 4, 1089, Budapest, Hungary
[2] Bioinformatics Center, Institute for Chemical Research, Kyoto University, Gokasho, 611-0011, Uji, Kyoto, Japan
[3] Research Unit for Realization of Sustainable Society, Kyoto University, Gokasho, 611-0011, Uji, Kyoto, Japan
[4] Hanoi University of Science and Technology, Dai Co Viet Road No. 1, 100000, Hanoi, Viet Nam
[5] Department of Computer Science, Aalto University, 02150, Espoo, Finland

Corresponding author: Petschner, Peter (petschner.peter@semmelweis.hu)

Current Opinion in Systems Biology 2025, **42**:100551

This review comes from a themed issue on **ICSB 2022**

Edited by **Matteo Barberis**

For complete overview of the section, please refer the article collection - ICSB 2022

Available online 18 June 2025

https://doi.org/10.1016/j.coisb.2025.100551

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Introduction

Drug–drug interaction (DDI) means that the effects of a drug changes in the presence of another, often resulting in unwanted adverse events (AEs) for patients. A recent meta-analysis in geriatric outpatient populations estimated the pooled prevalence of potential DDIs at 28.8%; however, with substantial heterogeneity [1]. Investigating actual DDIs, Dechanont et al. estimated a 1.1% median prevalence rate of DDIs upon hospital admissions [2]. In hospitalized patients in Ethiopia, 72.2% of patients may have had potential DDIs [3]. Among hospitalized patients in general, potential DDI prevalence ranged from 16.3% to 71.1% with a pooled rate of 33.1% excluding intensive care unit patients, where it was higher [4]. Excluding studies with poor quality increased the rate to 44% [4]. While actual patient harm was only registered in a few studies and was moderate [4], these results suggest that DDIs represent a significant health burden in the general population and, especially, in elderly, hospitalized patients and intensive care units.

The extent of polypharmacy, the use of multiple drugs in parallel, increases the risk of DDIs [5]. For the most common interactions related to the drug-metabolizing cytochrome P450 (CYP) enzymes, patients aged over 65 years and taking 5–9 medications already had a probability of 50% for a DDI, with the probability steeply increasing with the number of drugs [6]. Polypharmacy became more common between 1988 and 2010 [7]. Among the elderly, the average number of medications taken elevated from 2 to 4 and the number of people taking more than 5 medications increased 2.4 times until 2010 [7]. The temporal trends in polypharmacy and the associated elevated risk of DDIs suggest that their prevalence will become higher in the future and call for interventions to reduce the caused harm.

The first step in avoiding patient harm would be the identification of potential DDIs [8]. After identification, the DDIs can be avoided due to usually available alternatives in medication regimens and therapeutic guidelines. Given all these evidences and trends, the reliable identification of potential DDIs is imperative to reduce patient suffering and healthcare costs, especially in countries with aging populations.

The identification of potential DDIs can be formulated as a prediction problem for machine learning (ML), where the task is to predict new potential interactions from known DDIs. The increasing use of ML across a

variety of domains during drug development prompted the Food and Drug Administration (FDA) to recognize the potential of ML in all phases of drug development, including DDI discovery [9]. An ML-based approach can be considered low-cost and capable to recover previously unseen patterns from DDI data and thus, can be a key for reliable DDI identification.

Accordingly, numerous ML models have been proposed. For an earlier review about DDI prediction also discussing non−graph-based methods, see Ref. [10]. In general, supervised ML approaches, which include neural networks, tailor their internal structures during learning to achieve minimal prediction loss on the training data, i.e. the difference between the known value of a given sample and its prediction. The process allows, for example, neural networks to extract 'rules,' how the input (e.g. a vector) can be transformed into the output (e.g. a scalar). The training process can be so efficient that the algorithm may overfit to the training data, meaning low loss on training but high loss on unseen data. As usually the objective is to learn generalizable patterns, performance is tested on another, previously unseen, test dataset from the same sample to avoid overfitting. After optimizing and balancing the fit to the training and generalization to test DDI data, the trained model will be capable to predict DDIs through the application of the learned 'rules.' All in all, the aforementioned framework allows to identify DDIs that do not follow directly from the already established human knowledge of the drug, like off-target effects.

While quite a few non−graph-based, supervised models have been proposed regarding DDI prediction, these cannot directly utilize a common form of DDI and drug-molecule data graphs. Graphs contain the dependency structure between nodes in addition to node features and, thus, substantial information about node relationships, which can be utilized in DDI prediction.

Based on the aforementioned points, our focus in the present primer is to discuss 1) the graphs used in DDI prediction methods, 2) recent graph- and hypergraph-based methods for DDI prediction, 3) the used data sources in this setting, and 4) the future potential for the combination of more developed algorithms with large-scale protein and metabolite data to obtain more interpretable and reliable DDIs.

Types of graphs and applied machine learning methods on graphs in drug−drug interaction prediction

Most current models utilize two basic graphs for DDI prediction, which differ in their data content and, therefore, what they represent. The two types are as follows:

1. Molecular graphs:
    1. G = (V, E, A), where
    2. V is the *set of atoms in a drug* molecule
    3. E $\subseteq$ V x V is the set of *edges representing chemical bonds* between two atoms
    4. A is a matrix of *atom attributes/features* (e.g. molecular weight, charge, etc.), A $\in \mathbb{R}^{|V| x K}$, with K being the number of node features

The task is to learn a function $f(G_i, G_j, W) \rightarrow \mathbb{R}$, where G_i, G_j are molecular graphs of drug i, j, and W are learnable parameters, and the output is an interaction score for (d_i, d_j) drug pairs.

2. DDI graphs:
    1. G = (V, E, A), where
    2. V is the *set of drugs*
    3. E $\subseteq$ V x V is the set of *edges representing interaction* between two drugs
    4. A is a matrix of *drug attributes/features* (like protein binding characteristics, presence of particular substructures, etc.), A $\in \mathbb{R}^{|V| x K}$, with K being the number of node features

The task is to learn a function $f(G, (d_i, d_j), W) \rightarrow \mathbb{R}$, where W are learnable parameters, d_i, $d_j \in V$, and the output is an interaction score for (d_i,d_j).

Modifications to the aforementioned points have been proposed.

1. If there are more than one edge type, then G = (V, E, A, L) with L $\subseteq \mathbb{R}^{|E| x S}$ S being the number of edge types/labels. For example, in a molecular graph: type of bonds, or in a DDI graph: type of AEs. Such graphs are often called multirelational graphs or multigraphs.
2. If the set of nodes contains different type of nodes, for example $V = V_d \cup V_p \ldots \cup V_x$, where additional nodes are proteins, genes, diseases, etc. These networks are called heterogeneous, multimodal, or bioentity graphs. (Of note, multimodal graphs are multirelational too in the reviewed methods.)

Graph neural networks (GNNs) are deep learning methods that can be utilized to learn on graphs. In the DDI literature, almost exclusively convolutional graph neural networks (CGNs) are used to learn on molecular or DDI graphs (for readers with an interest on methodological details, we refer to Ref. [11]). CGNs can be divided into spatial and spectral CGNs. In spatial CGNs, the next layers' features are usually calculated by using the direct neighborhood of the node, defined by $N(v_i) = \{v_j \in V \mid (v_i, v_j) \in E\}$ [11]. Spectral convolution,

in contrast, originates in spectral graph theory and utilizes the graph Fourier transform of a signal (i.e. node features) and a learnable filter after the eigendecomposition of the graph Laplacian [11]. While the two approaches differ in their implementation, their idea is based on the assumption that exploitation of the graph topology information can lead to gains in prediction performance and, eventually, more accurate DDI predictions compared to non–graph-based methods. In case of molecular graphs, this topology represents chemical topology within molecules, while in case of DDI graphs, it is the DDI structure.

In the followings, we review some of the recent advances of DDI prediction algorithms using GNNs.

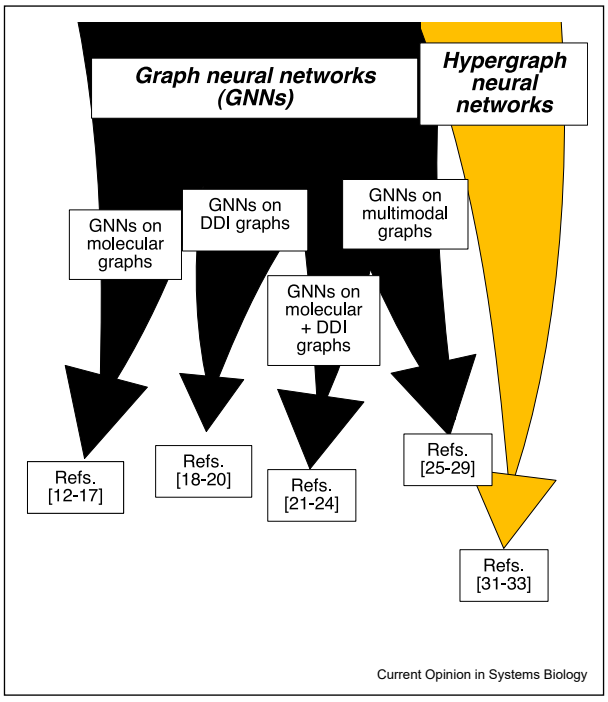
Recent advances with graph neural networks for drug–drug interaction prediction

The main directions in GNN-based DDI prediction is shown in Figure 1. We divide the methods based on which type of graphs is used as the GNN input.

Graph neural networks on molecular graphs

Among the methods utilizing GNNs exclusively on molecular graphs, Chen et al. used a bond type–aware message propagation CGN. In this work, node representations were learned through aggregation of neighboring nodes with the same bond type in a molecular graph, which contained multiple bond types. Attention-based graph pooling was used to get the final embedding of the molecular graphs, and weighted circular correlation was used to obtain scores for DDIs [12]. Feng et al. concatenated outputs of all layers of a graph attention network to extract hidden features of the drug molecular graph with different abstraction levels and fed the obtained features to a feed-forward neural network for interaction classification [13]. Another approach of Yang et al. worked with a CGN using bond type as edge features with substructure attention mechanism to obtain a score for each substructure and use it to calculate substructure–substructure interaction between molecules [14]. (Of note, substructure is not equivalent to chemical/pharmaceutical substructures. In ML, substructures are usually the learned subgraphs of the molecular graphs.) From a molecular perspective, the often used graph representation of molecules lacks explicit 3D information. This problem was addressed by He et al. with 3D chemical structure–aware message passing CGN layers in their method called 3DGT-DDI [15]. Ma et al. used bond type as edge features in the CGN on the molecular graph. Graph representations were retrieved from the different layers, and substructures were weighted based on their contribution to DDIs by a learnable parameter. DDI scores were calculated between the substructures of the drugs for every substructure and DDI type using a learnable weight matrix

Figure 1



Categorization of graph neural network–related research directions of drug–drug interaction prediction problem
Black arrows show major directions of graph neural network development based on the utilized graph type. Molecular graph–based methods use atoms as nodes and bonds between atoms as edges; drug–drug interaction (DDI) graphs use drugs as nodes and interaction between drugs as edges. Multimodal graphs contain drugs and additional node types (like proteins, genes, diseases, pharmacological groups, etc.) and the connections between these (thus, such graphs are multirelational). The orange arrow depicts hypergraph neural network development as a related field recently used for DDI prediction, where edges can contain more than two nodes (node types). Corresponding references are also indicated.

[16]. A recent method using 3D molecular graph information was Meta3D-DDI, where the authors used 3D molecular structure graphs, aggregated atom-level information, ensured invariance in 3D, and calculated a DDI score using the obtained representations [17].

Graph neural networks on drug–drug interaction graphs

Among the methods using GNNs only on homogeneous DDI graphs, Feeney et al. proposed an algorithm to increase CGN efficiency on large multirelational DDI graphs by sampling edges around a target edge in a type-dependent way and use the samples to update node features [18]. Kang et al. utilized a homogeneous DDI graph as a CGN input and combined all layers' output with a layer-wise fusing module to use representations of the DDI graph topology at different levels [19]. Al-Rabeah et al. learned on a DDI graph using drug substructure, pathway, enzyme, and target similarities as features of nodes [20].

Graph neural networks on molecular and drug–drug interaction graphs

Some of the methods used CGNs on both molecular graphs and homogeneous DDI graph. Wang et al. used spatial CGNs in a 'graph-of-graphs' setting to learn DDIs with GoGNN's [21]. Bai et al. also considered a similar setting and used graph attention networks [22]. He et al. used a gating mechanism with a graph attention network to regulate information flow from neighboring nodes in drug molecular graphs and learn their representations. Additionally, an extraction module on the drugs' Simplified Molecular Input Line Entry System (SMILES) string, encoding chemical structural information, was also used. Subsequently, the molecular graph representations and the SMILES latent features were used together in a DDI graph to obtain prediction scores [23]. Xie et al. inferred DDIs using a classifier module that was fed with embeddings calculated from a combination of molecular graph and DDI graph embeddings plus additional similarity matrices (Jaccard similarity) of drug, enzyme, and target similarities for the drug pairs [24].

Graph neural networks used on multimodal graphs

Some methods utilized multimodal networks, which contain other nodes in addition to drugs to utilize detailed information about potential biological/pharmaceutical entities, through which DDIs may occur.

Zitnik el al. developed Decagon, which predicts protein–protein, drug–protein, and DDI edges in a heterogeneous network in parallel. Decagon input consisted protein–protein interactions (PPIs), drug–protein interactions, and DDIs in a graph with 964 different DDI edges. The proposed architecture learned PPIs, DDIs, and drug–protein targets by using a CGN encoder and a tensor factorization decoder. Through this approach, Decagon could integrate PPI, drug-target, and DDI data with large performance gains compared to preceding methods [25].

Drug knowledge graphs are multimodal graphs but do not contain edges between drugs, only between drugs and biological entities, and are used for drug-relevant feature extraction for DDI prediction. Lyu et al. used a drug knowledge graph with drug-related protein bindings, like drug-target, drug-transporter, drug-enzyme bindings, and used a spatial convolution–based CGN on this graph to extract drug representations. In addition, molecular information in the form of structural, target, and enzyme feature similarity was also used, and a neural network calculated the interaction score for DDIs from drug representations and the similarities. The method, in essence, leveraged the learning of drug representations on drug knowledge graphs [26]. Another work used the GNN-learned representations of the DDI-relevant, pruned subgraphs from a drug knowledge graph, which included 11 different node types (genes, diseases, pathways, etc.). Molecular fingerprints together with the learned subgraph representations were used to calculate DDI score. The importance of the approach is related to the efficient extraction of DDI-relevant information from drug knowledge graphs and that the pruned subgraphs provided reasoning paths for the interpretation of the learned DDIs [27]. Han et al. extracted topological information from drug knowledge graphs using knowledge graph neural network (KGNN) and retrieved molecular information in the form of SMILES strings to obtain both chemical and topological latent features for drugs and used the dot product of the features of the two drugs to calculate a DDI score. It demonstrated the value of combining molecular and other types of data [28]. In attention-based cross domain graph neural network (ACDGNN), a heterogeneous, multirelational network containing drugs, genes, diseases, etc. (10 different node types were used) and its connections was utilized. The authors used cross-domain transformation to embed the different node types into the same space and propagated feature information from the different neighboring node types via an attention mechanism. Eventually, DDI scores were obtained through tensor factorization. ACDGNN's strength is the adaptation of cross-domain transformation paving the way for the use of large, complex, heterogeneous, multirelational DDI graphs in DDI prediction [29].

The aforementioned methods using heterogeneous graphs provided opportunities to use large amount of heterogeneous graph data for the DDI prediction task. Nonetheless, there is a more intuitive representation for both molecules and DDIs, hypergraphs.

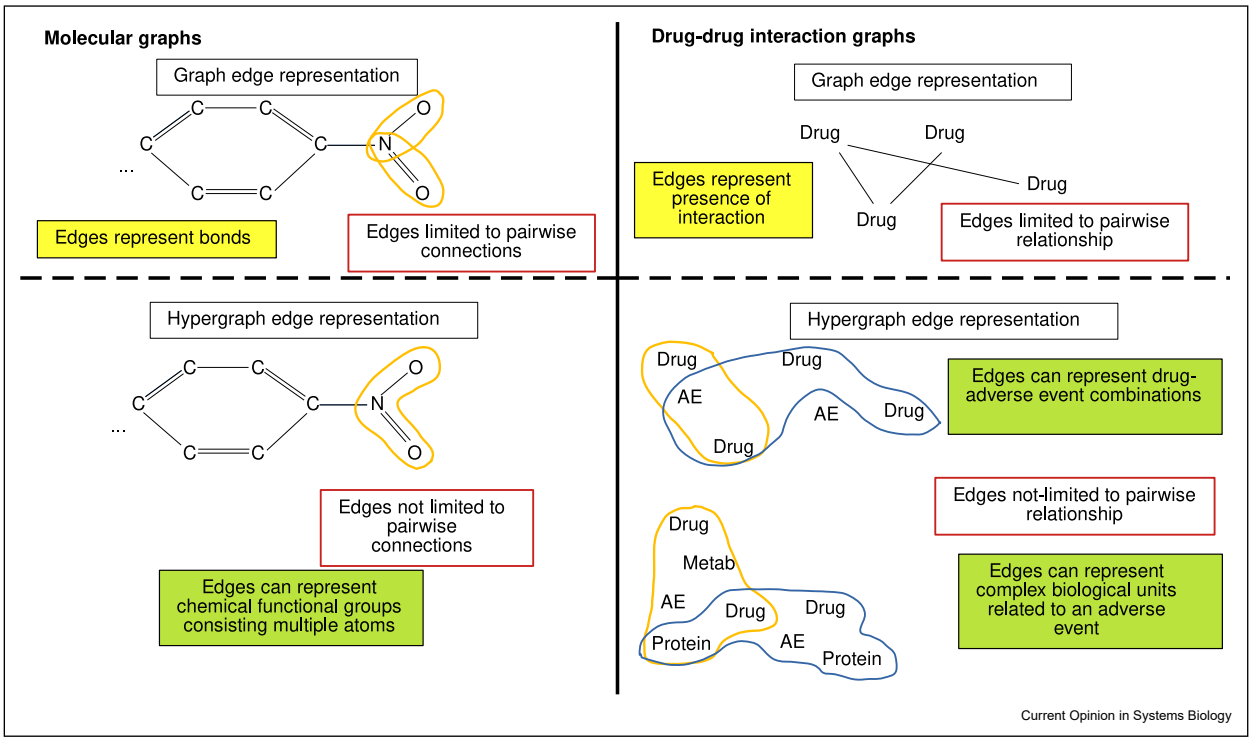
Hypergraphs as intuitive representations for drug–drug interactions

Hypergraphs are a more general concept than graphs, and their application gained momentum recently for DDI prediction. In a hypergraph, edges can contain more than two nodes [30], which provide conceptual benefits in comparison to molecular and DDI graphs, as shown in Figure 2. For the DDI prediction task, hypergraphs are more intuitive because 1) they can model polypharmacy interactions, when more than two drugs cause an interaction, 2) through the inclusion of AEs as nodes in the model, the relationship between AEs can also be utilized during training, and 3) heterogeneous nodes, including proteins, drugs, and AEs, can be efficiently modeled after projection. Despite these straightforward advantages, due to data and computational limitations, hypergraph neural network utilization to predict DDIs is currently limited.

Hypergraph-based drug–drug interaction prediction frameworks

Nguyen et al. defined DDIs in a hypergraph with both AEs and drugs being nodes, and hyperedges were

Figure 2



Conceptual benefits of using hypergraph representations in drug–drug interaction prediction tasks
The figure depicts conceptual benefits of hypergraph representations when 1) using molecular graphs, where edges can combine different atoms forming, e.g. chemical functional groups in a hypergraph instead of a simple bond or 2) when using DDI or multimodal graphs, where edges can allow for drug–drug–AE combinations or functional units of drug–protein–(metabolite)–AE combinations. (Edge colors denote different edge types in the examples.) Abbreviations: AE, adverse event, Metab, metabolite.

drug–drug–AE triplets. On this hypergraph, drug and AE representations were learned using a central smoothing assumption and spectral convolution, where the central smoothing assumption is that AE is close to the midpoint of the two drugs in the latent space. Both molecular structure- and protein-binding features were used for learning [31]. In another work, Nguyen et al. defined a similar hypergraph and learned both drug and AE latent features and an interaction matrix of such latent features to predict DDIs. In the work, the authors also implemented a sparsity assumption on the interaction matrix to represent sparsity of the DDI space (i.e. the ratio of reported DDIs compared to all potential drug–drug–AE triplets) [32]. Saifuddin et al. utilized hypergraphs based on the molecular substructure of the drugs and considered each drug as a hyperedge containing the nodes representing the substructures. Learning was performed through node and hyperedge attention networks [33].

In the aforementioned papers, the methods outperformed graph-based methods (see individual papers for comparisons and for validation of the identified DDIs, like in Ref. [32]), suggesting that hypergraphs are not only a more intuitive representation of the DDI problem but that their use allows for further development in terms of prediction performance.

Data sources for drug–drug interactions

In addition to algorithm development, another key to obtain more reliable DDI predictions both in graph- and hypergraph-based algorithms is high-quality DDI data.

A rich source of DDI data comes from postmarketing surveillance. DDIs are collected, assessed, and reported during clinical phases, and important DDIs are added to the drug leaflets [34]. However, before market authorization, drugs are tested only in a carefully selected human sample, while individual patient factors, like comorbid diseases, age, ethnicity, etc. can substantially influence drug efficacy, safety, and DDIs [35]. Therefore, during postmarket surveillance, additional data about AEs are collected in drug AE reporting systems (pharmacovigilance databases) in large populations and real-life settings according to the World Health Organization's (WHO's) Programme for International Drug Monitoring [36]. Reported cases in pharmacovigilance databases contain limited patient factors, concomitant

medications, diseases, and additional drug use—related factors in standardized format. Such databases are rich and useful sources of information for DDIs after quality filtering, particularly, because they contain reports about novel, previously unknown AEs with a higher chance.

TWOSIDES is a freely accessible, high-quality dataset in English language, generated from a pharmacovigilance database [37]. The original dataset was based on reports submitted to the FDA Adverse Event Reporting System between the first quarter of 2004 and the first quarter of 2009 [37]. Since then, TWOSIDES is regularly updated, and the new version of the dataset is accessible at [38], albeit most ML applications still use the old version. A drawback of pharmacovigilance databases is that individual differences [35], variable therapeutic guidelines, different drug availability, and the diverse reporting habits can influence which cases are reported in a given population. Thus, while TWOSIDES is a valuable resource for DDI prediction, additional data from other populations could help the discovery of more stable interactions across populations.

Another source of DDI information is DrugBank [39]. From databases, like DrugBank, extraction of AEs is not always trivial as these are represented as continuous texts requiring preprocessing in contrast to the standardized Medical Dictionary for Regulatory Activities (MedDRA) format of AEs in most pharmacovigilance databases. Nonetheless, due to the additional data contained in DrugBank, including, for example, protein binding information and chemical features for drugs, it can be a complementary source to TWOSIDES, ideal, e.g. to build multimodal graphs.

The Kyoto Encyclopedia of Genes and Genomes (KEGG) database can also be used to retrieve DDI information [40,41]. KEGG DRUG contains various data related to a given drug, including its DDIs and, importantly, the mechanisms of the DDI [42]. In contrast to DrugBank, which focuses only on drugs, KEGG also contains physiological processes in, e.g. metabolic pathways. Thus, KEGG is a data source for normal, physiological relationships and drug-related perturbations of the connections of biological entities. The rich information of KEGG is most often used to build multimodal graphs for the DDI prediction task, similarly to DrugBank.

According to the aforementioned points, the majority of cited ML algorithms used these resources to construct the graphs.

Additional data sources for drug–drug interaction prediction

DDIs are related to the drug's chemical properties, which seem to justify the inclusion of chemical structure descriptors or molecular graphs in the prediction task from sources, like PubChem [43]. However, chemical characteristics often have a complex and subtle relationship with the occurrence of interactions, mostly through the drug's binding characteristics. A more realistic view considers DDIs as final outcomes of complex interactions between drugs and a biological system; thus, it is beneficial to apply a system-based view of drug effects called systems pharmacology [44,45]. Multimodal, graph-/hypergraph-based approaches (e.g. Refs. [25—29,31,32]) can be especially suitable for such a perspective (Figure 2/b), but they require additional data sources more directly mapping drugs to their interacting partners in biological systems. As we argue in the following, protein and metabolic data are prime candidates to extend DDI graphs due to their direct connection with drug effects.

Protein data

Pharmacokinetics and pharmacodynamics are two branches of pharmacology, the discipline studying drug actions [46]. Broadly speaking, pharmacodynamics studies the drug actions on the organism, and pharmacokinetics deals with the actions of the organism on the drug [47]. The latter consists of 4 major processes: 1) absorption—the process of reaching the systemic circulation from the site of administration, 2) distribution—distribution of drug molecules in tissues, 3) metabolism—chemical structure modifications to prepare the molecule for excretion, and 4) excretion—removal of the original and modified drug molecules.

From a pharmacodynamic perspective, binding at macromolecule targets (protein, DNA, or RNA) is necessary for a drug to have an effect, suggesting that there should be on average at least one macromolecule for every drug [48,49]. With DNA- and RNA-binding drugs being rare, the macromolecule targets used by drugs are almost exclusively proteins. There are only few exceptions from the rule of macromolecule binding, like in case of antacids, osmotic and chelating agents, which exert effects relying directly on the chemical characteristics. However, even in the latter cases, protein binding cannot be excluded [50,51]. In fact, the reality may be far from a single target protein per drug. Prediction estimates indicated 329 protein targets on average per drug and 57 binding drugs per protein, with experimental evidence supporting drug promiscuity even across protein families [52,53]. Similarly, pharmacokinetic processes require crossing of membrane barriers (absorption, distribution, metabolism, and excretion), transportation in body fluids (distribution), and chemical modifications (metabolism), which involve proteins [54—56].

Alterations in protein-binding characteristics either by directly modulating the same protein or through PPIs are known to cause both pharmacodynamic and

pharmacokinetic DDIs [57]. Inclusion of protein binding data (and PPIs) can reflect these connections and allow learning on a more relevant graph.

In accordance, approaches using multimodal graphs often leveraged protein binding. In particular, Decagon utilized both protein-binding and PPI data and learned PPIs in parallel to DDIs in an effort to adopt a systems view [25]. In contrast to the sole use of chemical characteristics, the direct utilization of protein binding information could deliver more interpretable models too (e.g. in Ref. [32]) allowing to assess DDI-relevant proteins (of latent features) and feasibility evaluation of DDIs. In contrast, such inference is not possible through the use of chemical information alone.

A resource for protein binding data for DDI prediction is DrugBank, from which macromolecule associations of the drugs can be retrieved. The current version of DrugBank contains 19,395 drug-target, 5770 drug-enzyme, 3391 drug-transporter, and 936 drug-carrier associations (albeit with a small number of veterinary drugs included) [39]. Pathway-level data from KEGG may also be used to retrieve complex drug features related to proteins [42].

For PPI data, the search tool for the retrieval of interacting genes/proteins (STRING) is a prime resource with 296,567,750 high-confidence PPIs from 14,094 organisms [58], with human-specific data also available for download. In addition, the data of ref. [25] can also be utilized to use PPIs in algorithms.

Taken together, protein-binding and PPI data help to put DDIs into a systems context and allow for the interpretation of the interactions between a drug pair and the biological system.

Metabolite data

Metabolites are low-molecular-weight chemical compounds that are substrates and products of metabolism in biological entities [59]. The definition spans a wide range of molecules from active neurotransmitters to intermediate, metabolic by-products (e.g. drug metabolism) and confers tens of thousands of entities.

One way to incorporate metabolite information in DDI prediction is to focus on drug metabolites since a large share of drugs are exposed to both hepatic and nonhepatic metabolism [60,61].

The result of drug metabolism is a set of metabolites with the potential to contribute to or be responsible for both pharmacodynamic and pharmacokinetic drug metabolite−driven DDIs.

For a pharmacodynamic example, codein is extensively metabolized into active compounds, including morphine, by CYP2D6 and CYP3A4 enzymes. The resulting metabolites can participate in, e.g. interactions at the opioid receptors with other binding substances (fentanyl, oxycodon, etc.). In fact, numerous morphine derivatives are known to be metabolized into active compounds with the potential for similar DDIs [62].

Pharmacokinetic examples are also available. Interaction between sulfinpyrazone and warfarin was attributed to the capability of sulfinpyrazone to inhibit a hepatic enzyme, CYP2C9, involved in the metabolism of warfarin. However, a later study showed that a sulfide metabolite of sulfinpyrazone can inhibit the enzyme, while the parent compound is unlikely to be involved [63]. CYP3A4 is a key hepatic enzyme responsible for at least 30% of the metabolism of clinically used drugs, and its inhibition is known to cause DDIs [64]. In case of itraconazole, an antifungal medication with known CYP3A4-related DDIs, 50% of the inhibition of CYP3A4 after administration could be attributed to its 3 metabolites [65]. In fact, for more than 30 drugs were inhibitory metabolites proposed showing *in vitro* inhibition of one or more CYP enzymes [66]. While such mechanisms are understudied [66], new drug metabolite candidates for DDIs and better characterization of their potential interactions are regularly published [67,68].

Given the aforementioned points, it would be a promising direction for future studies to use drug metabolite data either as drug features or to expand drugs into a network of drug metabolites in the DDI graph. However, to our knowledge, no method tried to incorporate drug metabolite information into ML-based DDI prediction.

From a methodological perspective, inclusion of drug metabolites into a GNN-based DDI prediction task is challenging and should consider 1) how to integrate such data (e.g. drug metabolites are nodes, features, etc.), 2) meaning of edges as connections (e.g. multi-relational graphs), 3) latent representation of the different nodes (if drug metabolites and drugs are both nodes), and 4) interpretation of the DDIs, if needed.

From a data perspective, in addition to DrugBank, a rich source of information related to (drug) metabolites could be the Human Metabolome Database (HMDB) [69]. HMDB 5.0 contains more than 250,000 metabolites including 909 drug-derived ones. Data about the drug metabolite chemical structure and, in some cases, associated diseases, proteins, and ontology data can be retrieved [69]. Nonetheless, drug-related data in HMDB are incomplete. For example, in a recent study, 36 metabolites of tramadol were detected in the plasma and blood of an overdose case [70], but the HMDB lists only 10 results related to the drug. Probably, novel compounds will be added as untargeted metabolomic

investigations after drug use, which systematically measure metabolites in a patient, become more common. Despite lacking complete information, the HMDB can be considered as a currently available top resource for metabolites and could already be utilized to add drug metabolite data to ML-based predictions.

Potential integration of drug–drug interaction, metabolomic, and protein data

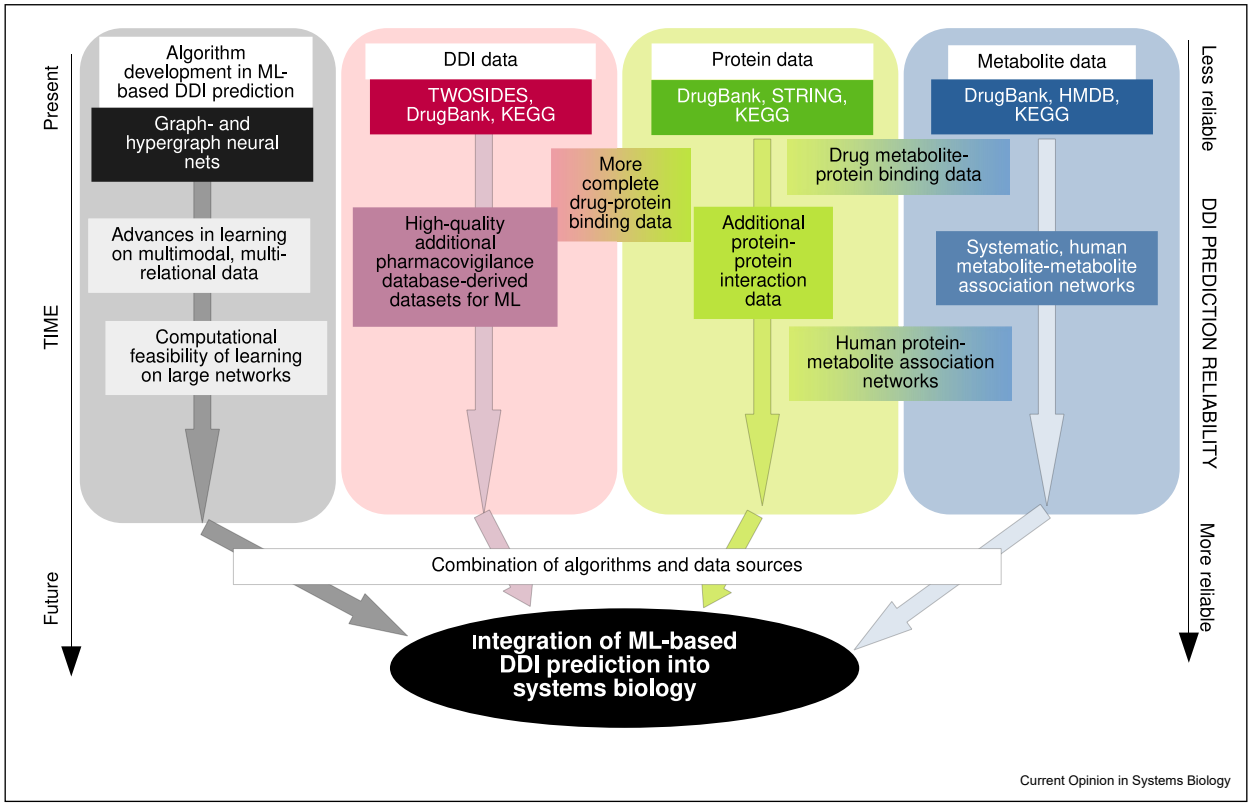
A future approach could use endogenous metabolites from metabolomic studies or KEGG instead of drug metabolites, shifting the perspective from drugs to the biological system. The extensive and combined use of physiological metabolite data with other information 1) would allow for the combination of DDI, PPI, metabolite–protein interaction [71], and metabolite–metabolite association networks [72–74], 2) would consider DDIs as perturbations of biological systems, and 3) could provide information about both 'normal' and 'drug' states. In this complex, multirelational and heterogeneous data structure connections between entities could be learned in parallel. Such approaches would combine systems pharmacology focusing on drug effects and systems biology focusing on the biological organism through ML (Figure 3).

Further perspectives

In addition to including metabolite data, there are further limitations that need to be addressed in future works.

Current methods favor high-level, generalized representations of drug characteristics, for example, by using yes/no variables for drug–protein binding instead of binding affinities, which could provide quantitative information. The case is similar for the output, meaning often a binary decision on whether two drugs interact or not, instead of, for example, dose-dependent probabilities. Both GNNs and hypergraph neural networks would

Figure 3



A potential future trajectory of physiological protein and metabolite data collection and machine learning (ML) algorithm development leading to gains in drug–drug interaction prediction

The figure shows how protein and metabolite data collection and ML algorithm development can integrate drug–drug interaction (DDI), protein, and metabolite data in an ML application that leverages information from all these sources. Inclusion of drug metabolites still confers a systems pharmacology–based view focusing on drugs and is already used in the present algorithms. Extension of the physiological protein–protein interactions with metabolite–metabolite information, however, facilitates a systems biology point of view. Combining these two perspectives with powerful ML models considers both perturbed and physiological states and could infer information about both. HMDB, Human Metabolome Database; KEGG, Kyoto Encyclopedia of Genes and Genomes, STRING, the search tool for the retrieval of interacting genes/proteins.

be capable to use and predict quantitative variables, which may represent a potential future direction.

In addition, human physiology is also treated in an abstract sense, assuming that there is no individual variation, i.e. DDIs are independent of the patient's, age, sex, etc. These issues can be traced back to available data quality and quantity, which do not allow predictions on individual levels. For example, KEGG and DrugBank do not contain individual-level data; therefore, these could not be used for individual-level predictions. Pharmacovigilance databases, in theory, offer case-level details; however, these only include age, sex, weight, and height of the individual and the diseases someone may have. Labor parameters, detailed descriptors, therapy errors, or any other characteristics that may influence DDIs are not recorded. Even for the already mentioned basic characteristics, like age and sex, data may be missing, especially in older records. In addition, databases are often maintained by regulatory agencies in separate countries, and while reporting may be standardized, reports are still recorded in individual languages, albeit there are attempts to mitigate this problem. The European Union collects such reports and aggregates them centrally [75], but individual-level reports remain hard to access. Another database, the Japanese Adverse Drug Event Report database contains records only in Japanese, limiting access to researchers familiar with the language [76]. VigiBase [77], the central resource of the WHO for drug AEs (and DDIs) is also only accessible freely on aggregate levels. These restrictions, while perfectly aligned with patient privacy, restrict the potential of ML. Novel, accessible, extended datasets containing, e.g. more accurate dosage regimens, should be made available in English or in large, integrated, individual-level databases to allow individual-level predictions. Inference would be possible without violating patient rights using decentralized datasets in federated learning paradigms [78] with GNNs already adapted to such settings [79].

Nonetheless, even in such cases, in general, population-specific and individual-specific data would be insufficient to interrogate 3-way, 4-way, etc. interactions, which occur when three, four, etc. drugs interact with each other because the sampling size compared to the possible number of potential interactions is very small. Additionally, high doses are only rarely administered to patients due to the already elevated risk of AEs and interactions, limiting prediction accuracy for these cases and the usefulness of predictions in high-risk clinical settings, like in intensive care units. Therefore, collection of additional data should be continuous, and these data sources should ideally be synthesized and freely available to mitigate the scarcity of data.

ML developers are also often faced with the hard task to quality control reporting data, which requires a deep understanding of pharmacovigilance databases. This may be one of the reasons, why developers mostly rely on already quality-controlled standardized databases, like TWOSIDES, on which algorithms may show increased performance but can fail to do so under real-life settings. Interdisciplinary research teams, involving pharmacologists, pharmacovigilance experts, and ML developers and the adoption and deep integration of ML, especially graph- and hypergraph-based methods into the DDI recovery pipeline during drug development, may help alleviate these problems.

Currently, the distinction between pharmacodynamic and pharmacokinetic DDIs remains unsolved in ML. There are methods that predict CYP enzyme binding [80] and GNNs were also used for the task [81]; however, categorizing DDIs into these two major types would require labeled data, i.e. a DDI dataset in which interactions were categorized into pharmacokinetic and pharmacodynamic interactions to develop a supervised learning model. Until such data become available, interpretable algorithms, like [32] may be used to inform about the probable category of a potential DDI by highlighting possible underlying mechanisms.

Conclusions

In the present primer, we elucidated the current state of graph-based DDI prediction by reviewing the recent literature on the field and expressing major development directions of the algorithms. Graph-based methods are certainly promising, and hypergraph-based methods also began to appear on the field showing superior performance. While methodological development is ongoing, the inclusion of data sources in addition to DDI data could further boost algorithm performance and medical relevance of found DDIs. Protein binding data were already shown to deliver more interpretable and biologically useful results, while the benefit of metabolite data could complement and, thereby, advance these approaches. Metabolite data currently suffer from incompleteness; however, the utilization of drug metabolite information is already within reach. In the future, complex ML models combining data of relationships between and within drugs, metabolites, and proteins can be envisioned, which could allow to infer not only new DDIs but also new protein and metabolite connections. In addition, deeper embedding of ML into discovering DDIs during drug development and better, individual data collection and dissemination strategies could improve prediction performance and clinical utility. Such approaches would deeply integrate DDI data through its interactions with biological entities into systems biology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could

have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by Japan Society for the Promotion of Science (Postdoctoral Fellowships for Research in Japan, standard program) [P20809] to P.P., MEXT KAKENHI [19H04169, 20F20809, 21H05027, 22H03645 and 25H01144] and Novel Applications of Artificial Intelligence in Physical Sciences and Engineering Research (AIPSE) program by Academy of Finland to H.M.

Data availability

No data was used for the research described in the article.

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